Air Astana Group | Aircraft Availability - Research Memo

Problem

Air Astana Group (including FlyArystan) disclosed 22 unscheduled PW1100G engine removals in 2025, with up to 13 aircraft grounded in peak season. The Group estimated a USD 42.3 million adverse impact on EBITDAR from unscheduled engine removals. [S1, S3, S4]

Process bottleneck

Accelerated engine removals feed into inspection and MRO queues; long off-wing periods and constrained replacement-engine supply can leave otherwise usable aircraft unavailable. Industry-wide context is reported by Simple Flying. [S1, S2]

Four distinct potential causes

1. Mandatory/accelerated engine removals; 2. limited repair capacity and shop turnaround; 3. limited available spare engines; 4. peak-season capacity pressure and fleet scheduling constraints. These are proposed mechanisms, not individually measured causal effects. [S1-S4]

Null hypotheses and method

H0-1: Mean modeled downtime is equal in peak and off-peak seasons. H0-2: Mean modeled downtime is equal when spare engines are available versus unavailable. Welch two-sample t-tests on 1,500 entirely synthetic observations.

Results - synthetic scenarios only

Peak 11.27 h versus off-peak 6.48 h (difference 4.79 h; p=9.52e-33). Spare available 5.28 h versus unavailable 10.22 h (difference 4.94 h; p=1.41e-44). Reject both H0s within the designed simulation ONLY; no empirical effects on Air Astana are established.

Recommendations

Evaluate the Group-reported mitigation measures: leasing spare engines and additional A320-family aircraft, dynamic capacity management, and faster MRO turnaround. [S4] Pilot an aircraft-on-ground dashboard and test the actual effects using anonymized maintenance, inventory and scheduling logs before investing.

Source integrity

S1 official Air Astana 2025 annual report; S2 Simple Flying (17 Sep 2026); S3 The Engine Cowl (13 Mar 2026); S4 Simple Flying (13 Mar 2026). Four publications across three publishers, but S3/S4 largely relay company disclosures and S2/S4 share a publisher. Full article texts S2-S4 supplied by user; external site retrieval not independently verified. Synthetic data are NOT real flight or maintenance records.